

INTRODUCTION TO SYNTAX I

Session 1: Language has (hierarchical) structure

Timea Szarvas

University of Potsdam

`timea.szarvas@uni-potsdam.de`

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QUICK ROUND OF INTRODUCTIONS

- What's your name?
- What's your native language?
- Rate your knowledge of formal syntax:



TODAY'S PLAN

- ◉ The role of syntax in the language faculty
- ◉ What makes the syntax we do formal?
- ◉ Basic building blocks
- ◉ Hierarchy
- ◉ Terminology and ways to represent hierarchical structure

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- ⊙ We unwrap linguistic utterances to understand the thoughts behind them

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■ How do we do that?

- ⊙ By following a set of rules: form-meaning relations are not arbitrary
- ⊙ We obey these rules effortlessly and in real time (at least in our native language)

■ Following the rules of your native language

- ⊙ We do not think about these rules or have the ability to state them explicitly

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- ⊙ A speaker of English knows *that* (1b) is ill-formed, but cannot explain *why*:¹

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⇒ *Tacit knowledge*: complete, permanent, implicit, not consciously acquired

■ The goal of formal linguistics

- ◉ Make this tacit knowledge explicit to come up with a theoretical model of it.

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LANGUAGE FACULTY: Y-MODEL

Mental lexicon

■ **Mental lexicon**

Contains all functional and lexical elements of the language L.

LANGUAGE FACULTY: Y-MODEL

Mental lexicon



Numeration
[likes, Mia, cats]

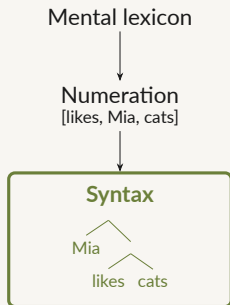
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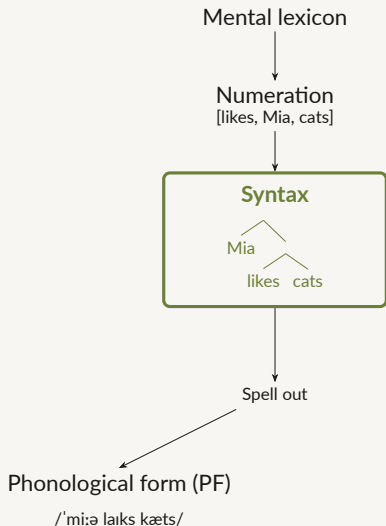
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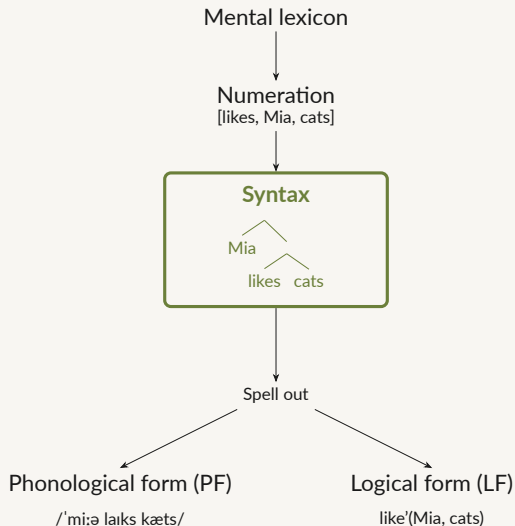
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■ Phonological form

Determines phonetic interpretation.

■ Logical form

Determines semantic interpretation.

WELCOME TO SYNTAX!

■ Syntax is the computational component of mental grammar.

- ① **Creativity** – the number of complex structures that can be formed is infinite.
- ② **Restrictiveness** – the manner in which complex structures are formed is non-arbitrary.

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■ Central questions:

- ⊙ What are the smallest building blocks from which linguistic structures can be formed?
- ⊙ What are the rules that determine how these building blocks are combined?
- ⊙ How are these rules ordered? To what extent do these rules vary cross-linguistically?

■ Natural languages are diverse.

- ◉ All possible orders of S(ubject), V(erb) and O(bject) are attested:
- ◉ **SOV**: Turkish (Turkic)
- ◉ **SVO**: BCMS^a (Slavic)
- ◉ **VSO**: Irish (Celtic)
- ◉ **VOS**: Malagasy (Malayo-Polynesian)
- ◉ **OVS**: Urarina (isolate, Peru)
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⇒ There is no general rule in syntax prohibiting either of these orders — which order is dominant is acquired on a language-by-language basis.

■ Some languages do not have a dominant word order at all, like Hungarian (Ugric)!

- (2) Anyu el-mond-ott egy vicc-et.
mom PV-tell-PST a joke-ACC
S V O
'Mom told a joke.'

CROSS-LINGUISTIC VARIABILITY

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(3) a. Anyu egy viccet elmondott.
S O V
b. Egy viccet elmondott anyu.
O V S

(4) a. Egy viccet anyu elmondott.
O S V
b. Elmondott anyu egy viccet.
V S O
c. Elmondott egy viccet anyu.
V O S

- ⊙ 'No dominant word order' does not mean that the orders are interchangeable in every context – it means that they are equally acceptable and meaningful in the language.
- ⊙ The notation in the second row of (2) is called the *glossing* or the *glosses*. We use them to make linguistics structures legible to anyone regardless of their proficiency of the respective language. The glosses used in this class follow the Leipzig Glossing Rules.

■ Other phenomena are globally restricted.

- ⊙ No language allows questioning an object of a coordinate structure.

- (5) a. Betty is reading Nietzsche.
b. What is Betty reading ____ ?
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(6) *Hungarian*

*Mi-t olvas Betti [Nietzsche-t és ____]?

what-ACC reads Betti Nietzsche-ACC and

Intended: Which x is such that Betty is reading Nietzsche and x?

Try (5d) in your native language — awful, right?

■ How do we uncover syntactic rules if speakers are unable to state them?

- ⊙ Directly observable data (language that is spontaneously produced), only tell us what *is* possible (e.g. Hungarian word order), but they do not reveal what *is not* (e.g. coordinate island constraint)

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- ⊙ So called **parasitic gaps** are infrequent, yet most speakers find them quite good (in English).
- ⊙ Negative evidence is crucial – data showing that something is not possible (recall how bad those questions formed from coordinate structures were in (5d) and (6)!)

■ **Solution: Collect judgments about what is and isn't possible.**

- (8) a. *What is Betty reading Nietzsche and ____ ?
b. *Who did you invite Hannah and ____ ?
c. *Which advanced class would Sam skip the intro and ____ ?

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■ Stability of judgments

- ⊙ No matter which words we plug into the sentence and no matter which speaker we ask, questioning an object from a coordinate structure is impossible.

I Observation:

- ⊙ Recognize **regularities** in linguistic data and **formulate hypotheses**
- ⇒ *Questioning objects from coordinate structures is bad.*

BUILDING A THEORETICAL MODEL

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No conjunct or element from it may be *extracted* (= questioned).

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I Testing:

- ⊙ Test whether the **empirical predictions** are correct
- ⇒ *Is questioning an object from any part of a coordination bad across languages?*

A NOTE ON THEORETICAL MODELS

“Scientists also make golems.² Our golems rarely have physical form, but they too are often made of clay, living in silicon as computer code. These golems are scientific models. But these golems have real effects on the world, through the predictions they make and the intuitions they challenge or inspire. A concern with ‘truth’ enlivens these models, but just like a golem or a modern robot, scientific models are neither true nor false, neither prophets nor charlatans. Rather they are constructs engineered for some purpose.” (McElreath 2020)

²An animated, artificial human figure made from inanimate matter, such as clay or mud, originating in Jewish folklore.

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- Models attempt to depict the observed reality as accurately and simply as possible.
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■ What you learn about in this class is a particular theoretical model of human syntax.

- ⊙ It is called Minimalism (based on the minimalist program by Chomsky 1995).
- ⊙ I draw some inspiration from David Adger's 2003 book *Core Syntax* (Adger 2003).

BUILDING A THEORETICAL MODEL

- (9) a. Betty is reading Nietzsche and Tom is citing Kant.
b. *What is Betty reading Nietzsche and Tom citing ____ ?

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c. *What is Betty reading ____ and Tom citing Kant?
d. What is Betty reading ____ and Tom citing ____ ?
e. What is Betty reading ____, Tom citing ____ and Stella studying ____ ?

Revision: ATB-rule (ATB = across-the-board, Ross 1967)

- ⊙ Questioning the object of a coordination is bad *unless* it affects every conjunct.

Next questions:

- ⊙ What does the syntactic structure of these sentences look like?
- ⊙ (testing + revision repeated endlessly until our model covers the empirical facts)

(10) *The cats chases the mouse.

(11) ?The mouse that the cat that the dog barked at chased panted.

■ Both sentences are not quite right. Can you guess why?

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- ⊙ (9) is syntactically ill-formed: our **linguistic competence** tells us that verbs agree with their subjects in English.
 - ⊙ (10) is syntactically well-formed, but really difficult to understand.
- ⇒ Our **linguistic performance** is failing us! Center embedding is terrible to process.

■ In German, center embedding for relative clauses is optional.

- (12) a. [?]Die Maus, die die Katze, die der Hund anbell-te, jag-te, schnauf-te.
 DET mouse REL DET cat REL DET dog bark.at-PST.3SG hunt-PST.3SG pant-PST.3SG

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⇒ The fact that the syntactically equivalent (11b) is an improvement illustrates that the oddness of (11a) is not due to grammar/competence.

⇒ **Linguistic competence** = the tacit knowledge linguists are interested in.

⇒ **Linguistic performance** = competence in action, limited by general cognitive constraints.

■ Linguistic competence:

- ⊙ The abstract set of rules of a language L that generates grammatical structures.
 - ⊙ Grammaticality is **categorical**: the system only generates grammatical structures.
- ⇒ A complex expression is grammatical or ungrammatical, no in between.

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■ Linguistic performance:

- ⊙ The real-life application of linguistic competence.
 - ⊙ Acceptability is **gradient**: numerous factors influence acceptability (plausibility, context, processing ease, prescriptivism, **grammaticality!**)
- ⇒ A complex expression can be more or less acceptable, the nuances are unlimited.

■ Linguistic structures are built from lexical and functional elements.

- ⊙ Intuitively, some building blocks belong together 'more closely' than others.

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■ Can you find more sensible units in (13)?

- ◉ These units are called *constituents*.
 - ◉ Constituents may consist of multiple words, but each word is also a constituent.
 - ◉ Several grammatical processes affect constituents, not random strings.
- ⇒ We can use these grammatical processes to test whether a string is a constituent!

CONSTITUENCY TESTS

- ① Replacement test $\Rightarrow$ constituents can be replaced
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(17) It was **in the cabinet** that the mouse that stole the cheese was hiding from the cat.

- ⑤ Coordination test $\Rightarrow$ constituents can be coordinated

(18) The mouse that stole the cheese was hiding from **the cat** and **the dog** in the cabinet.

■ **Note: the inference drawn from these tests is unidirectional!**

- ⊙ If a chain α can be replaced/elided/moved/... $\Rightarrow \alpha$ is a constituent.
- ⊙ **NOT:** If a chain α is a constituent, it must be possible to replace/elide/move/...it.

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■ Independent reasons may lead to a failed test.

- ⊙ The tests make use of syntactic operations (movement, ellipsis, coordination) that have their own restrictions and quirks.
- $\Rightarrow$ Always apply as many constituency tests as possible!

EVIDENCE FOR HIERARCHICAL STRUCTURE

■ There is something special about our previous example:

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- ◉ (12'): *the cat in the cabinet* forms a constituent
- ◉ (12''): *hiding in the cabinet* forms a constituent (without *the cat*)

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- ⊙ Y-model = formalization of the hypothesis that mental grammar is modular
 - ⊙ The output of syntax is mapped onto Phonological form (PF) and Logical form (LF).
 - ⊙ LF can only work with the input syntax has passed on to it!
- ⇒ **Structural ambiguity = multiple possible LFs = multiple syntactic structures.**

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■ This argument has the same logic as the one we saw before.

- ⊙ When *Chinese dishes* forms a constituent, stress shifts (*chiNESE* $\Rightarrow$ *CH*inese)!
- ⊙ PF can only work with the input syntax has passed on to it.
- ⊙ **Multiple PFs = multiple syntactic structures.**

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HIERARCHY AND TREES IN MATHEMATICS

■ Putting the *formal* in *formal syntax*

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■ Borrowing representations from mathematics

- ◉ We already formalized the structure of a sentence by adding brackets:

(12') The mouse that stole the cheese was hiding from [the cat [in [the cabinet]]]

(22) $4 + [6 \times 2]$

I Borrowing more representations from mathematics

- ⊙ Syntax is all about trees — but where do these trees actually come from?

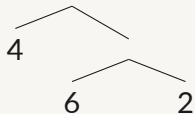
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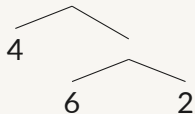


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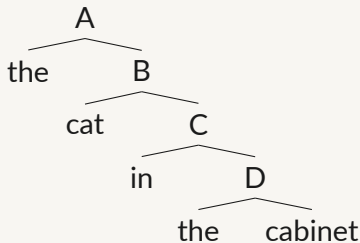
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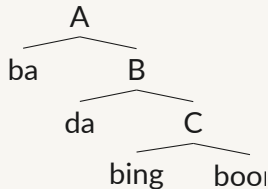
(24) [the cat [in [the cabinet]]]



LET'S TALK ABOUT TREES!

Terms denoting parts of/relations within a tree:

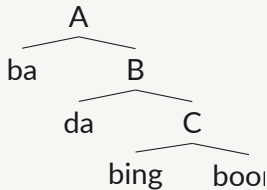
- Nodes: *A, B, C, ba, da, bing, boom*



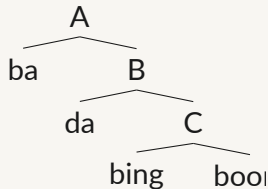
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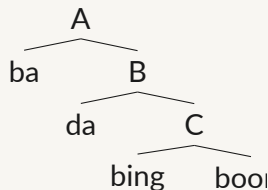
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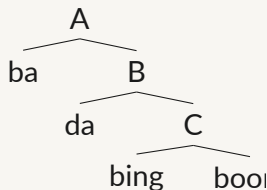
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- ◉ **Dominance:** A dominates *ba*, B and everything dominated by B (B dominates *da*, C and everything dominated by C, C dominates *bing* and *boom*)

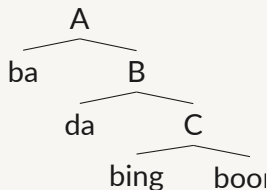
LET'S TALK ABOUT TREES!



Terms denoting parts of/reasons within a tree:

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- ◉ **Terminal nodes:** *ba*, *da*, *bing*, *boom*
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- ◉ **Immediate dominance:** A immediately dominates *ba* and B but not *da*, C, *bing* and *boom*

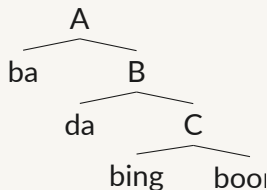
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- ◉ **Motherhood:** A is the mother of *ba* and B, B is the mother of *da* and C, C is the mother of *bing* and *boom*
- ◉ **Sisterhood:** *ba* and B, *da* and C, *bing* and *boom* are sisters

■ Some useful definitions to return to if you feel lost:

- ◉ **Node:** represents a constituent of a tree
- ◉ **Branch:** connects the nodes of a tree
- ◉ **Terminal node:** a node that does not branch any further
- ◉ **Path:** continuous sequence of nodes and branches that connects two nodes
- ◉ **Root:** the topmost node dominating all other nodes
- ◉ **Dominance:** node α dominates node β iff³ α is higher up the tree than β such that you can trace a line from α to β going only downwards.
- ◉ **Immediate dominance:** node α immediately dominates node β iff it is the closest node that dominates β .
- ◉ **Motherhood:** node α is the mother of node β iff it immediately dominates β (node β is the daughter of node α).
- ◉ **Sisterhood:** nodes β and γ are sisters iff node α is the mother of β and γ .

³if and only if

Ask away!

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